

Assignment 2

Mapper Class:-

```
package org.serverlog;
```

```
import java.io.IOException;
```

```
import org.apache.hadoop.io.IntWritable;  
import org.apache.hadoop.io.LongWritable;  
import org.apache.hadoop.io.Text;  
import org.apache.hadoop.mapred.*;
```

```
public class log_mapper extends MapReduceBase implements Mapper<LongWritable,  
Text, Text, IntWritable> {  
    private final static IntWritable one = new IntWritable(1);  
  
    public void map(LongWritable key, Text value, OutputCollector<Text, IntWritable>  
output, Reporter reporter) throws IOException {  
  
        String valueString = value.toString();  
        String[] map_Data = valueString.split("-");  
        output.collect(new Text(map_Data[0]), one);  
    }  
}
```

Reducer Class:-

```
package org.serverlog;
```

```
import java.io.IOException;  
import java.util.*;
```

```
import org.apache.hadoop.io.IntWritable;  
import org.apache.hadoop.io.Text;  
import org.apache.hadoop.mapred.*;
```

```
public class log_reducer extends MapReduceBase implements Reducer<Text, IntWritable,  
Text, IntWritable> {  
  
    public void reduce(Text t_key, Iterator<IntWritable> values,  
OutputCollector<Text,IntWritable> output, Reporter reporter) throws IOException {  
        Text key = t_key;  
        int sum = 0;  
        while (values.hasNext()) {  
            // replace type of value with the actual type of our value  
            IntWritable value = (IntWritable) values.next();  
            sum += value.get();  
        }  
        output.collect(key, new IntWritable(sum));  
    }  
}
```

```
}
```

Runner Class:-

```
package org.serverlog;
```

```
import org.apache.hadoop.fs.Path;
```

```
import org.apache.hadoop.io.*;
```

```
import org.apache.hadoop.mapred.*;
```

```
public class log_analysis_main {
```

```
    public static void main(String[] args) {
```

```
        JobClient my_client = new JobClient();
```

```
        // Create a configuration object for the job
```

```
        JobConf job_conf = new JobConf(log_analysis_main.class);
```

```
        // Set a name of the Job
```

```
        job_conf.setJobName("log_analysis_main");
```

```
        // Specify data type of output key and value
```

```
        job_conf.setOutputKeyClass(Text.class);
```

```
        job_conf.setOutputValueClass(IntWritable.class);
```

```
        // Specify names of Mapper and Reducer Class
```

```
        job_conf.setMapperClass(org.serverlog.log_mapper.class);
```

```
        job_conf.setReducerClass(org.serverlog.log_reducer.class);
```

```
        // Specify formats of the data type of Input and output
```

```
        job_conf.setInputFormat(TextInputFormat.class);
```

```
        job_conf.setOutputFormat(TextOutputFormat.class);
```

```
        // Set input and output directories using command line arguments,
```

```
        //arg[0] = name of input directory on HDFS, and arg[1] = name of output  
directory to be created to store the output file.
```

```
        FileInputFormat.setInputPaths(job_conf, new Path(args[0]));
```

```
        FileOutputFormat.setOutputPath(job_conf, new Path(args[1]));
```

```
        my_client.setConf(job_conf);
```

```
        try {
```

```
            // Run the job
```

```
            JobClient.runJob(job_conf);
```

```
        } catch (Exception e) {
```

```
            e.printStackTrace();
```

```
        }
```

```
    }
```

```
}
```

Output:-

```
10.1.1.236    7
```

```
10.1.181.142    14
```

10.1.232.31	5
10.10.55.142	14
10.102.101.66	1
10.103.184.104	1
10.103.190.81	53
10.103.63.29	1
10.104.73.51	1
10.105.160.183	1
10.108.91.151	1
10.109.21.76	1
10.11.131.40	1
10.111.71.20	8
10.112.227.184	6
10.114.74.30	1
10.115.118.78	1
10.117.224.230	1
10.117.76.22	12
10.118.19.97	1
10.118.250.30	7
10.119.117.132	23
10.119.33.245	1
10.119.74.120	1
10.12.113.198	2
10.12.219.30	1
10.120.165.113	1
10.120.207.127	4
10.123.124.47	1
10.123.35.235	1
10.124.148.99	1
10.124.155.234	1
10.126.161.13	7
10.127.162.239	1
10.128.11.75	10
10.13.42.232	1
10.130.195.163	8
10.130.70.80	1
10.131.163.73	1
10.131.209.116	5
10.132.19.125	2
10.133.222.184	12
10.134.110.196	13
10.134.242.87	1
10.136.84.60	5
10.14.2.86	8
10.14.4.151	2
10.140.139.116	1
10.140.141.1	9
10.140.67.116	1
10.141.221.57	5
10.142.203.173	7

10.143.126.177	32
10.144.147.8	1
10.15.208.56	1
10.15.23.44	13
10.150.212.239	14
10.150.227.16	1
10.150.24.40	13
10.152.195.138	8
10.153.23.63	2
10.153.239.5	25
10.155.95.124	9
10.156.152.9	1
10.157.176.158	1
10.164.130.155	1
10.164.49.105	8
10.164.95.122	10
10.165.106.173	14
10.167.1.145	19
10.169.158.88	1
10.170.178.53	1
10.171.104.4	1
10.172.169.53	18
10.174.246.84	3
10.175.149.65	1
10.175.204.125	15
10.177.216.164	6
10.179.107.170	2
10.181.38.207	13
10.181.87.221	1
10.185.152.140	1
10.186.56.126	16
10.186.56.183	1
10.187.129.140	6
10.187.177.220	1
10.187.212.83	1
10.187.28.68	1
10.19.226.186	2
10.190.174.142	10
10.190.41.42	5
10.191.172.11	1
10.193.116.91	1
10.194.174.4	7
10.198.138.192	1
10.199.103.248	2
10.199.189.15	1
10.2.202.135	1
10.200.184.212	1
10.200.237.222	1
10.200.9.128	2
10.203.194.139	10

10.205.72.238	2
10.206.108.96	2
10.206.175.236	1
10.206.73.206	7
10.207.190.45	17
10.208.38.46	1
10.208.49.216	4
10.209.18.39	9
10.209.54.187	3
10.211.47.159	10
10.212.122.173	1
10.213.181.38	7
10.214.35.48	1
10.215.222.114	1
10.216.113.172	48
10.216.134.214	1
10.216.227.195	16
10.217.151.145	10
10.217.32.16	1
10.218.16.176	8
10.22.108.103	4
10.220.112.1	34
10.221.40.89	5
10.221.62.23	13
10.222.246.34	1
10.223.157.186	11
10.225.137.152	1
10.225.234.46	1
10.226.130.133	1
10.229.60.23	1
10.230.191.135	6
10.231.55.231	1
10.234.15.156	1
10.236.231.63	1
10.238.230.235	1
10.239.100.52	1
10.239.52.68	4
10.24.150.4	5
10.24.67.131	13
10.240.144.183	15
10.240.170.50	1
10.241.107.75	1
10.241.9.187	1
10.243.51.109	5
10.244.166.195	5
10.245.208.15	20
10.246.151.162	3
10.247.111.104	9
10.247.175.65	1
10.247.229.13	1

10.248.24.219	1
10.248.36.117	3
10.249.130.132	3
10.25.132.238	2
10.25.44.247	6
10.250.166.232	1
10.27.134.23	1
10.30.164.32	1
10.30.47.170	8
10.31.225.14	7
10.32.138.48	11
10.32.247.175	4
10.32.55.216	12
10.33.181.9	8
10.34.233.107	1
10.36.200.176	1
10.39.45.70	2
10.39.94.109	4
10.4.59.153	1
10.4.79.47	15
10.41.170.233	9
10.41.40.17	1
10.42.208.60	1
10.43.81.13	1
10.46.190.95	10
10.48.81.158	5
10.5.132.217	1
10.5.148.29	1
10.50.226.223	9
10.50.41.216	3
10.52.161.126	1
10.53.58.58	1
10.54.242.54	10
10.54.49.229	1
10.56.48.40	16
10.59.42.194	11
10.6.238.124	6
10.61.147.24	1
10.61.161.218	1
10.61.23.77	8
10.61.232.147	3
10.62.78.165	2
10.63.233.249	7
10.64.224.191	13
10.66.208.82	2
10.69.20.85	26
10.70.105.238	1
10.70.238.46	6
10.72.137.86	6
10.72.208.27	1

10.73.134.9	4
10.73.238.200	1
10.73.60.200	1
10.73.64.91	1
10.74.218.123	1
10.75.116.199	1
10.76.143.30	1
10.76.68.178	16
10.78.95.24	8
10.80.10.131	10
10.80.215.116	17
10.81.134.180	1
10.82.30.199	63
10.82.64.235	1
10.84.236.242	1
10.87.209.46	1
10.87.88.214	1
10.88.204.177	1
10.89.178.62	1
10.89.244.42	1
10.94.196.42	1
10.95.136.211	4
10.95.232.88	1
10.98.156.141	1
10.99.228.224	1